

Complexity and “Closure to Efficient Cause”

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Abstract

This paper has two main purposes. First, it will provide an introductory discussion of hyperset theory, and show that it is useful for modeling complex systems. Second, it will use hyperset theory to analyze Robert Rosen’s metabolism-repair systems and his claim that living things are closed to efficient cause. It will also briefly compare closure to efficient cause to two other understandings of autonomy, operational closure and catalytic closure.

Introduction

Toward the end of his book *Life Itself*, Robert Rosen (1991; see also Rosen 2000) claimed that the defining feature of living systems is that they are “closed to efficient cause”. The idea of closure to efficient cause is wrapped up with a series of other concepts (metabolism-repair systems, computability, impredicativity, complexity) in a somewhat obscure way. (See Letelier, Soto-Andrade, Abarzúa, Cornish-Bowden and Cárdenas 2006 for a recent attempt at clarification.) The first purpose of this paper is to set out exactly how these concepts are related. The upshot of this is that closure to efficient causation is a variety of complexity, where complex systems are systems whose models contain impredicativities, and, therefore, are not computable. Because of this complexity, the mathematical and computational tools of the mainstream cognitive sciences, derived from computability theory and designed for simple systems, are inappropriate as tools for analyzing systems that are closed to efficient causation (e.g., metabolism-repair systems). So, the second, more important, purpose of this paper is to introduce non-well founded set theory or hyperset theory (Aczel 1988, Barwise and Etchemendy 1987, Barwise and Moss 1996, Kerckel 2003), and show that it is useful tool to make sense of the kind of complexity that is characteristic of living systems.

Our plan is as follows. We will begin by introducing non-well founded set theory or hyperset theory. Hyperset theory comes with an intuitive, easy-to-understand system of graphs that make it possible to test systems for complexity. In the next section, we will discuss hypersets and introduce the graphing system. We will show that complex systems will all have graphs with a similar structure—they all have loops. In the following section, we will use these tools to analyze Robert Rosen’s metabolism-repair systems (1991). The purpose of this

will be to show that, as suggested by Kerckel (2003), the diagrams do indeed capture the complexity of Rosen’s model. Furthermore, it will allow us to say what distinguishes systems that are closed to efficient cause from other complex systems. This will allow us to briefly discuss the relationship among closure to efficient cause and two other proposed base cases of autonomous system: operationally closed systems (Varela 1979) and catalytically closed systems (Kauffman 1993). Finally, we very briefly discuss further work we have done in applying hyperset theory to complex systems and our plans for future work. We have already used hypersets to model the Belousov-Zhabotinsky reaction and animal-environment systems of the kind studied by ecological psychologists. This work suggests that hyperset graphs can be used as “complexity detectors” throughout the sciences, useful for determining just which systems are complex and in what ways (Chemero and Turvey 2006). In the future, we intend to use hypersets to develop a general theory of autonomy and context dependence.

Introducing Hyperset Theory

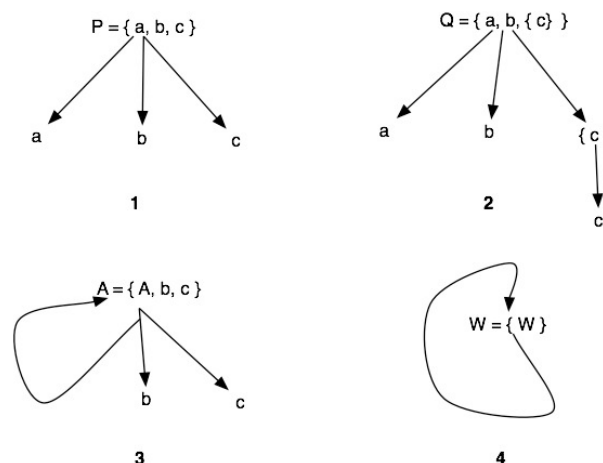
During the late 19th and early 20th century work that codified and established set theory, it became evident that certain paradoxes threatened the foundations of set theory. The most famous of these paradoxes was first discussed by Bertrand Russell, and is now known as *Russell’s Paradox* (1903). Imagine the barber who shaves all and only those who don’t shave themselves. Who shaves the barber? It turns out that if the barber shaves himself, he does not shave himself, and vice versa. Put in terms of sets, the paradox is as follows. Call all the sets that do not contain themselves as members *normal sets*; those that do contain themselves are *abnormal sets*. Consider S the set of all normal sets. Is S normal? If so, S is a member of the set of all normal sets. But since S is the set of all normal sets, S would then be abnormal. If S is abnormal, then S is a member of itself. But if S is a member of itself (the set of all normal sets), then S must be normal. So: S is a member of itself just in case it is not a member of itself, or, equivalently, S is normal just in case it is abnormal. This means that the sentence “S is a normal set” is simultaneously true and false. (We ask that readers convince themselves of this before continuing.)

This was taken to be a serious problem with the theory of sets, which was intended to be deducible from logic. In

response to this problem, Russell introduced the *vicious circle principle*, which outlawed any sentence which was circular. This was later formalized as the theory of types, most of the details of which we do not need to be concerned with, but whose main effect was to make it illegal for sets to be members of themselves. According to the theory of types, that is, there are no abnormal sets, and there is also no set of all sets. Paradox solved. Poincaré's (1906) banishment of impredicative definitions from mathematics, which may be more familiar to some readers, serves a similar purpose. Roughly, a predicative definition applies to members of some domain so that its application is not altered by addition of new members to the domain; an impredicative definition, on the other hand, picks out different members of a domain should new individuals be added to the domain. The definition that picks out "the individuals in Ms. Riley's class whose first names begins with P" is predicative, but the definition that picks out "the 4th tallest individual in Ms. Riley's class" is impredicative. A new student's arrival does not change the first letter of Peter's name, but it may make him 5th tallest. Essentially, impredicative definitions pick out individuals or properties whose falling under that definition depend on other members of a set. Another way to put this is that impredicative definitions pick out individuals in a way that is context-dependent. "Abnormal set" and "the barber who shaves all and only those who do not shave themselves" are defined impredicatively, so outlawing impredicativities makes defining these types of set impossible. Again, paradox solved.

The elimination of abnormal sets leaves us with standard set theory as taught in College-level logic and mathematics courses. This set theory is usually referred to as *well-founded*. There are two related senses in which standard set theory is well-founded. First, it is well-founded in that its implementation does not lead to paradoxes; that is, it is logically well-founded. The second way in which it is well-founded comes from graph theory, a standard tool used to understand set membership. In graph theory, every set determines a tree-like graph in which the set itself is the upper-most node and each member of the set is a descendent node with an arrow pointing to it from the original set of which it is a member; this is applied recursively. Figure 1 shows a graph for the set $P = \{ a, b, c \}$. Notice that there is an arrow from the node representing the set P to each of its members. Figure 2 shows a graph for the set $Q = \{ a, b, \{ c \} \}$. Notice that in this graph, there is an additional complexity in which the node for the member of Q, $\{ c \}$, has an additional member to its member the object c. The sets P and Q are well-founded sets; they are not abnormal and do not contain themselves as members. The graphs of P and Q show a property common to the graphs of all well-founded sets. Graphs of well-founded sets contain no infinite cycles or loops, and each path from the original set terminates in some object (its foundation, as it were). In fact, Aczel 1988 proves that a graph will contain no cycles or loops if and only if it is

well-founded. The 'if and only if' means that a graph that



Figures 1-4.

contains loops or cycles is a picture of a non-well-founded set. The presence of cycles and/or loops, that is, would indicate that some set has itself as a member, or that the concept, system, or definition it models is impredicative.

Consider, as an example, the abnormal set $A = \{ A, b, c \}$. This will have the graph in Figure 3. Aczel also defines the special abnormal set W as the set that contains only itself, $W = \{ W \}$. W is graphed in Figure 4. The sets A and W are not allowed in well-founded set theory, but they do have coherent and useful graphs. Aczel's Anti-Foundation Axiom, which embraces non-well-founded (i.e., abnormal) sets is as follows: every graph, loopy or not, pictures a genuine set. Hypersets are defined as graphable sets. So well-founded and non-well-founded sets are both types of hyperset.

Non-well-founded sets were banished from set theory in the interest of the logical soundness of the theory, and with the explicit goal of reducing mathematics to set theory, and set theory to logic. Re-introducing non-well-founded sets into our set theory makes such reduction impossible. But this loss (if it really is a loss) is offset by the increased ability to model real-world phenomena. Work by Gupta (1981), Barwise and Etchemendy (1987), Gupta and Belnap (1993) and Barwise and Moss (1996) makes very clear that many concepts and real-world systems are circular or otherwise not-well-founded, hence illegal according standard set theory. The most well-known example is the concept 'truth', and the predicate 'is true'. Gupta (1981) argues that the semantic paradoxes involving truth, such as the liar ("This very sentence is false") and the truth-teller ("This very sentence is true") show that the predicate "is true" can only be defined circularly, and develops a logical technique (the revision theory) for dealing with circular concepts. Later, Barwise and Etchemendy (1987) use hypersets to understand the truth predicate. Another example from Quine (1963) is "the most average Yale student". The identity of this individual

is only determined with respect to all the other current Yale students.

In each of these cases, some object in a system is defined impredicatively, and is therefore illegal in standard set theory. Of course, the illegality of these systems and concepts in standard set theory does not change their behavior, or cause them to cease to exist. Such systems do exist. Adequately modeling them requires a richer set theory, such as hyperset theory, that allows and accounts for the behavior of systems with impredicativities.

The graph theory that goes with hyperset theory will be especially important in what follows. As Aczel has proved and figures 1-4 demonstrate, graphs of hypersets can be used to determine whether the set pictured is or is not well-founded, and if it is not well-founded exactly what it is that makes it non-well-founded. These graphs, that is, can be used to determine whether systems contain impredicativities, and so are complex. The graphs also allow us to determine the precise source and nature of the complexity in such systems. We will use this feature of hypersets and their graphs to look at Rosen's model of living things as metabolism-repair systems.

Rosen on Life Itself

According to Rosen (1991, 2000), a central feature of living things is *complexity*. Rosen, it should be noted, has an idiosyncratic understanding of complexity. First, according to Rosen, whether a system is complex depends only secondarily on the system itself; it depends primarily on the system's models. A physical system is *simple* if and only if its behavior can be captured by a model that is Turing-computable; otherwise, the system is *complex*. So, for Rosen, a system is complex just in case its models are non-computable. This connects in vital ways with the above discussion of impredicative definitions and Russell's theory of types. Put most simply, Russell introduced the theory of types and Poincaré outlawed impredicative definitions specifically to keep logic, set theory and mathematics well-founded or, put anachronistically, Turing-computable. Thus systems whose models disobey the theory of types or contain impredicativities—models, that is, that are not well-founded—are complex in Rosen's sense. If this is so, we can use hyperset diagrams as a means of modeling systems that are complex. Indeed, we

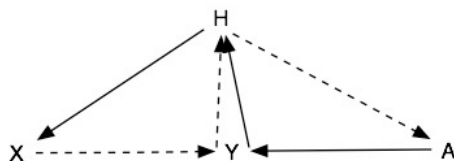


Figure 5. Rosen's model of living systems (1991). Note that $Y = H(X)$ is represented by a solid line from H to X and a dashed line from X to Y.

can use hyperset diagrams of models as tests to determine whether those models are complex.

We can see this by examining Rosen's metabolism-repair model of living systems (1991, 2000). According to Rosen, it is a necessary condition on living systems that their models contain three functions: a metabolic function, a repair function, and a replication function. We will discuss these in order. The metabolic function h_i is a mapping from raw materials $x_i \in X$ to behavior $y_i \in Y$. That is, it determines the way the system reacts to environmental input. A living system must also have a repair function that repairs the metabolic function to keep the system reacting appropriately to the environment. That is, it must have a function which maps behaviors $y_i \in Y$ onto the members of set H, where metabolic $h_i \in H$. Finally, the repair function must itself be repaired. This repair is done by the system's behavior. That is, each member y_i of the set of behaviors Y is itself a replication function that maintains the repair function, by mapping h_i onto $a_i \in A$. This is, admittedly, complicated. The idea,

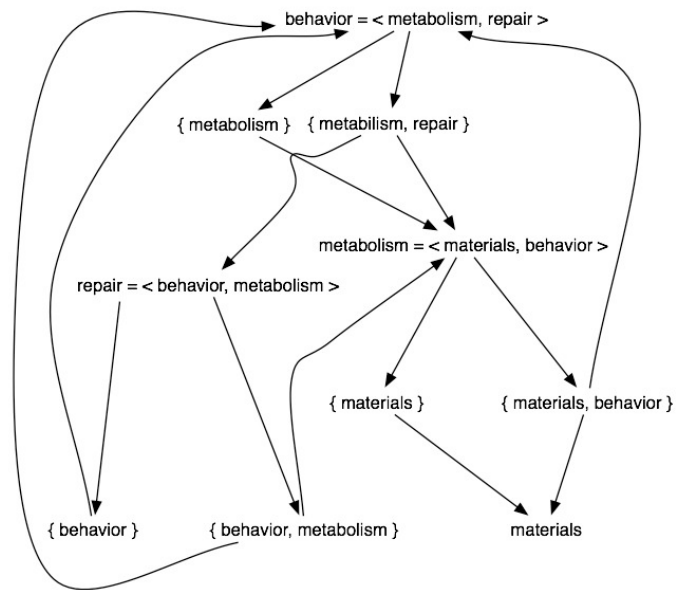


Figure 6. Simplified hyperset diagram of Rosen's metabolism-repair model of living systems. Note that functions are represented as ordered pairs composed of the domain and the range. That is, $a = f(b)$ is represented as $f = \langle a, b \rangle$.

though, is rather simple. The living system must be metabolizing consistently, so must maintain the causal processes that it uses in metabolizing. This is the job of the repair function. But the repair function must also be maintained, and this is done by the system's behavioral output. Rosen's diagrammatic representation of his model of living systems is shown in Figure 5. Note that in this diagram a function P from Q to R is represented as 'P' connected to 'Q' with a solid arrow, and 'Q' connected to 'R' with a dashed arrow.

Figure 6 is a hyperset diagram of Rosen's model of living systems. In figure 6, two common set-theoretic conventions are followed. First, functions are represented as ordered pairs. That is, if P is a function from Q to R , P is represented as $P = \langle Q, R \rangle$. Second, to preserve their ordering, ordered pairs are represented as sets containing the set containing their first member and the set containing both their members. Thus $\langle Q, R \rangle$ is represented as $\{ \{ Q \}, \{ Q, R \} \}$. In Figure 6, the sets of inputs X , behaviors Y , metabolic functions H , and repair functions A are represented as if they were individuals. This is done to make the already-complicated diagram simpler.

The thing to notice about the hyperset diagram in figure 6 is that there are loops, indicating that the sets are not well-founded. Indeed, in the graph there is only one location where the system does not loop back on itself: the set X of inputs from the environment. This makes sense of two important aspects of Rosen's model of life. First, the loopiness indicates that Rosen's diagram is indeed complex in that it involves irremovable impredicative loops. Rosen took just this kind of complexity to be one of the most important features of living systems, and often wrote of the impredicativities found in living systems. Figure 6 bears this out. Second, the fact that the only place the diagram "bottoms out" is with raw materials is also important. Rosen often wrote that living systems are "closed to efficient causation". We can see this in figure 5, Rosen's own representation of living systems. In that diagram, only X , the raw material input from the environment, is not both (1) itself a function in the system and (2) the output of a function of the system. To put this in Rosen's (and Aristotle's) language, all functions but X are both efficient and material causes in the system. X is only a material cause. The hyperset diagrams make clear how this is the case.

Other Definitions of Autonomy

With this analysis of Rosen's closure to efficient cause in hand, we can briefly consider two other analyses of autonomy: Varela's operational closure (1979) and Kauffman's catalytic closure (1993).

A system is operationally closed if and only if every effect of the system is also a part of the system. Varela, Thompson and Rosch (1991) offer the one-dimensional cellular automaton Bittorio as an example of an operationally closed system. Bittorio is operationally closed because the state of each of its cells has an effect only on the state of some of Bittorio's other cells—all these effects are within Bittorio. But note that Bittorio is not closed to efficient cause. Bittorio's operations (i.e., the rules that generate its behavior) are not produced by Bittorio, but are instead determined from outside, by a computer programmer. Figure 7 is a hyperset diagram of three of Bittorio's cells, which follows the same

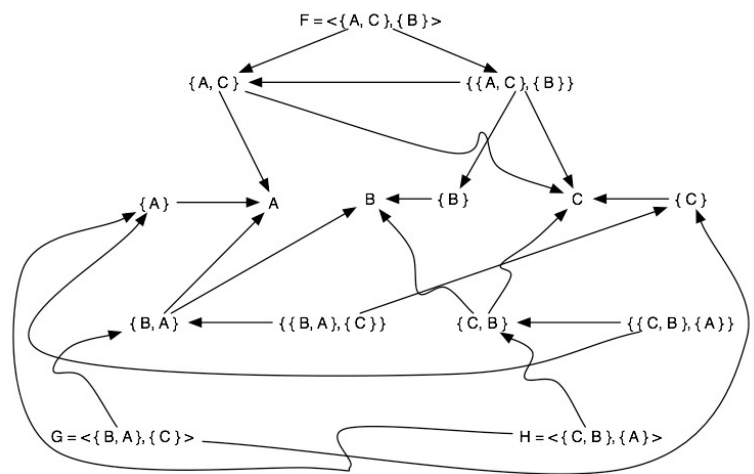


Figure 7. Three cells of the cellular automaton Bittorio. Each cell's state is a function of that of its neighbors. So $B = F(A, C)$. This is represented as $F = \langle \{A, C\}, \{B\} \rangle$.

conventions as figure 6. Figure 7 looks quite complicated, but it contains no impredicative loops: each of the system's functions (F , G , and H) is independent of the others.

This is not to say, however, that operational closure is not interestingly related to closure to efficient cause. A subclass of operationally closed systems, the autopoietic systems (Maturana, Varela, and Uribe 1974, Varela 1979) are closed to efficient cause. Autopoietic systems are systems that are self-producing. They are operationally closed because their effects are system-internal; in addition, though, the effects are also system-constituting. The autopoietic system most commonly discussed is the cell. In a cell, the metabolism produces a cell membrane and the membrane provides a boundary, which plays a crucial role in the cell's metabolism. This is depicted in Figure 8, which follows the same set theoretic convention as figures 6 and 7. Note that the diagram has a form very similar to Rosen's metabolism-repair system: every function in the system (the metabolism, the membrane) is produced by the system itself and only raw materials are not produced by the system. In other words, the diagram shows that autopoietic systems like the cell are closed to efficient cause.

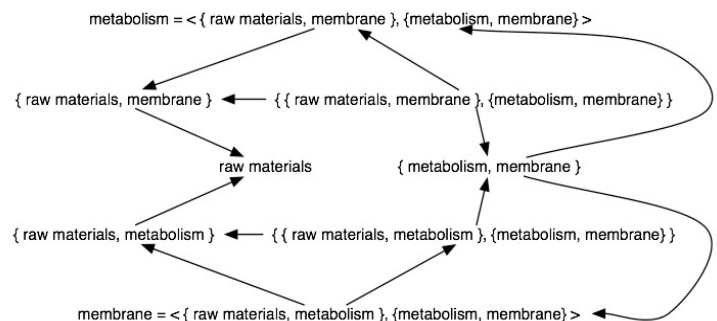


Figure 8. The cell and membrane system, a canonical autopoietic system.

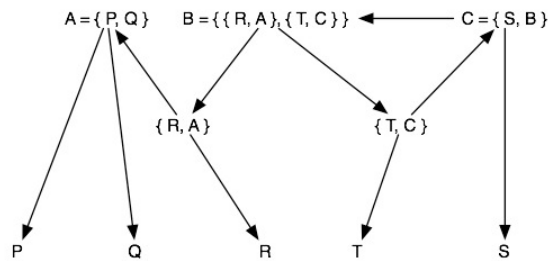
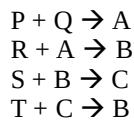


Figure 9. A hyperset representations of a catalytically closed set of reactions. Products are represented as the set of reactants that produce them.

Now consider another proposed minimal case of autonomy: catalytically closed systems (Kauffman 1993). A system is catalytically closed just in case every product of the system is also a catalyst in the system. As with operationally closed systems, not all catalytically closed systems will be closed to efficient cause. This is the case because although every product in a catalytically closed system is a catalyst for some reaction in the system, not every reaction need be catalyzed by a system product. If we put this in functional terms, in a system that is closed to efficient cause, every function is the output of some other function. In contrast, in a catalytically closed system, every output of every function is an input to some other function. We can see this in Figure 9, which is a diagram of the following (imaginary) catalytically closed set of reactions:



This set of reactions is catalytically closed (every product is a catalyst), but the first reaction is not catalyzed by a system product. In the hyperset diagram, we represent this system by equating each product with the set reactants that produce it. Thus the product A is $\{P, Q\}$; B is $\{\{R, A\}, \{T, C\}\}$; C is $\{S, B\}$. In the figure, there is an impredicative loop involving catalysts B and C. But A, which is a product of the system and a catalyst, is not involved in a loop. This shows that the system is a complex system (because it contains an impredicative loop), but it is not closed to efficient cause. Thus, in general, catalytically closed systems are not closed to efficient causation.

If we assume that closure to efficient cause is the most basic kind of autonomy (note that we have given no particular reason to do so), then autopoietic systems are autonomous, but most catalytically closed and operationally closed systems are not. This analysis agrees with Chu and Ho (2006), who dispute Rosen's purported proof that artificial life is impossible. Autopoietic systems, like metabolism-repair systems, are closed to efficient cause. Yet autopoietic systems are defined solely in terms

of their organization, not their material (Varela et al 1974; Varela 1979), so ought to be possible *in silico*. This implies that artificial systems that are closed to efficient cause are possible.

Conclusion

We take it that the hyperset analysis of Rosen's model of living systems is useful for three reasons. First, it demonstrates some of Rosen's claims about his model, viz., that it is non-computable, contains impredicativities, is closed to efficient causation. It also shows exactly what the relationship among parts of a system must be for it to be closed to efficient causation. We think, therefore, that the hyperset diagram is enlightening. Second, it allows us to see in some detail how closure to efficient cause differs from operational closure and catalytic closure. Third, and most importantly for our purposes, the hyperset diagrams show that such an analysis is useful and does in fact capture important properties concerning the complexity of models.

In other work on hypersets and complex systems (Chemero and Turvey 2006), we show that the hypersets can also provide illuminating models of complex systems like the autocatalytic Belousov-Zhabotinsky reaction and animal-environment systems as understood in Gibsonian ecological psychology. In future work, we intend to use hypersets to develop a theoretical account of cotenxt dependence.

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Working towards ‘basic autonomy’: a simulation model to study minimal self-producing cells

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Abstract

In this paper we propose a stochastic simulation model to investigate minimal ‘self-(re-)producing’ cellular systems. The problem is approached from a ‘bottom-up’ perspective, in the sense that we investigate how relatively simple cell components/processes can engage in a far-from-equilibrium dynamics, staying in those conditions by means of a rudimentary but effective control on the matter-energy flow through it. By means of a recently developed computational platform to carry out stochastic simulations of chemical reaction networks taking place in dynamic cellular compartments [Mavelli & Ruiz-Mirazo (2006)], we search for the first and most simple forms of *autonomy* in the context of lipid (more specifically, lipid-peptide) vesicles that enclose a proto-metabolic reaction network.

Introduction

Maturana and Varela’s work on autopoietic systems [Maturana & Varela 1973; Varela et al. 1974] was a cornerstone in the construction of a new way of thinking about biological systems, in which the concepts of organization and autonomy played a central role [Varela 1979]. And it also constituted a very important contribution to the development of Artificial Life as a discipline [Varela & Bourgine 1991].

However, most of their claims have somehow diffused away from the real practice of researchers in this field, and it is not clear whether their main message still holds or not. In this paper we retrieve the original autopoietic goal, looking into the problem with a rather different lens. Taking as a standpoint previous critical work on the theory of autopoiesis [Ruiz-Mirazo & Moreno 2004], we propose a simulation model to study the origins of minimal autonomous cells in more realistic conditions.

The work here reported is an elaboration on a model for minimal cell dynamics [Mavelli & Ruiz-Mirazo 2006], in which the self-assembly processes of simple lipid molecules were coupled with different reaction schemes for their synthesis, bringing about vesicles with interesting biophysical properties and time behaviours (homeostasis, growth, division...). Our point of departure (the ‘pre-autonomy’ scenario) will be a model cell that produces the lipid of which its membrane is made via an internal

reaction cycle. Then, we propose a more complete scheme to include amino acids and peptides (with their alleged autocatalytic properties [Kauffman 1986; 1993]) in the cell dynamics.

The final aim is to develop a model that can address the issue of how short peptide chains come to interact with the vesicles in a way that the latter may profit from the interaction to become more elaborate compartments (e.g., establishing small channels which would help them overcome problems like the accessibility of certain substrates to the interior, or the regulation of osmotic imbalances) and be able to host --in a truly autonomous way-- a proto-metabolic reaction network. Although it constitutes a rather simplified portrayal of the situation, we believe that this work might bring some light into the difficult but prebiotically very relevant question of how could simple cellular systems manage to couple internal chemical reactions with transport processes.

In any case, our approach is a step forward with regard to previous models of proto-metabolic cells (e.g.: [Varela et al. 1974; McMullin & Varela 1997; Dyson 1982; Csendes 1984; Fernando & Di Paolo 2004; Munteanu & Solé 2006; Ono & Ikegami 1999; Madina et al. 2003]) because it takes into account (in an abstract but, still, quite realistic way) the active role and dynamic properties of a cellular compartment, made of a membrane (lipid bilayer phase) and a water core. This is why we consider it well-suited to study the problem of the origins of autonomous cells.

The actual implementation of the model shares some features with Ganti’s ‘chemoton’ [Ganti 2002; 2003], but also shows important differences. For instance, diffusion and transport processes are here explicitly taken into account, as well as the possibility that the membrane changes its composition and functional properties. In addition, the original ‘template’ autocatalytic subsystem of the chemoton model is (a) removed or (b) substituted by a more general ‘polymer’ autocatalytic subsystem.

The major assumption behind this work is that lipid-peptide cells had --and were able-- to become minimal autonomous agents, solving basic problems like the control on concentration gradients, or the harvesting of